

HyperID: Ontology-Grounded Explainable Breed-Ancestry Estimation for Mixed and Indigenous Companion Animals — A Neuro-Symbolic Approach

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Abstract—For animal welfare platforms, shelters, and veterinary clinics, accurate classification of dog breeds is essential. Current CNN-based methods frequently fail to acquire supplementary information in a variety of circumstances, which results in poor recognition of difficult classes, particularly when there are visually comparable categories or out-of-distribution mixed and indigenous breeds. We present an ontology-grounded breed-ancestry estimation system that combines a multi-task neural network predicting physical traits with zero-shot Vision-Language Models (CLIP) through a log-space ensemble. In contrast to single-architecture black-box models, our method captures a variety of properties (Ear Shape, Coat Type, Coat Pattern, Size, Snout Length, Tail Carriage) and reasons over an OWL (Web Ontology Language) knowledge base. By recalibrating these fused features with SPARQL trait-overlap reasoning, our system creates a discriminative space for robust classification. In tests on the Oxford-IIIT Pet dataset subset, the system achieves 91.92% accuracy, competitive with traditional CNNs. Through ontology-based trait tracing, the model provides comprehensible conclusions, effectively identifies visually comparable classes, and demonstrates exceptional generalization to out-of-distribution mixed breeds (e.g., Goldador, Gerberian Shepsky) without re-training. These findings demonstrate that neuro-symbolic multi-branch architectures are a dependable and scalable solution for real-world environmental and animal welfare systems.

Index Terms—Neuro-Symbolic AI, Explainable AI (XAI), Knowledge Engineering, Ontology, Animal Welfare, Breed-Ancestry Estimation, Deep Learning, CLIP.

I. INTRODUCTION

Classifying animal breeds from images is crucial for everyday animal welfare operations, shelter management, and veterinary care. Accurate breed pattern recognition improves adoption matching systems, helps organizations organize their activities, and empowers veterinarians to make well-informed decisions about breed-specific health risks. Conventional image classification techniques are insufficient due to the complexity and lack of representation of mixed breeds and indigenous strays. Predicting ancestry accurately can save resources by averting misidentifications, lowering return rates for adoptions, and preventing behavioral mismatches. Current systems use a variety of deep CNN models to detect fixed purebred classes. Deep learning techniques might be highly

accurate on closed datasets, but ontology-grounded systems are more explainable and require less retraining, which makes them appropriate for dynamic real-world applications where new mixed phenotypes appear constantly.

Despite several approaches, there is still a knowledge gap regarding mixed-breed visual dynamics. We suggest a neuro-symbolic ensemble as a solution to this problem: For the classification of mixed-breed images, hybrid CLIP-Neural fusion with OWL Ontology reasoning emphasizes robust feature extraction, interpretability, and fine-grained recognition. While the neural ensemble improves discriminative capacity, the ontology reasoning ensures model explainability, validating decisions. This paper’s contributions are as follows:

- **Novel Hybrid Architecture:** We suggest a neuro-symbolic pipeline that combines continuous representations from CLIP and a Multi-Task Trait Network in a novel way. Log-space ensemble weighting is added to this combination to improve state-of-the-art accuracy by capturing complementing visual hierarchies.
- **Ontology Grounding:** To improve generalization and handle unseen classes, a thorough OWL ontology incorporating canonical breed traits and indigenous placeholder phenotypes is employed.
- **Attention-Enhanced Fusion:** In order to focus on discriminative information and reduce dataset bias, the ensemble adaptively recalibrates confidence scores from zero-shot textual prompts and neural visual heads prior to reasoning.
- **Explainability:** By highlighting trait-specific matches through SPARQL queries, the reasoner produces logic-based traces that offer an interpretable validation of the model’s decision-making process.

The remainder of the paper is organized as follows. Section II reviews relevant literature, Section III details the proposed methodology, including model architecture, training setup, and dataset preparation, Section IV presents and discusses experimental results, and Section V concludes the paper with future research directions.

II. LITERATURE REVIEW

Many studies have advanced automatic breed recognition using machine learning and deep learning for early and accurate identification. Researchers have applied DenseNet and ResNet with transfer learning on the Stanford Dogs dataset, achieving over 95% accuracy, though requiring large datasets and careful tuning. Other works applied GoogLeNet pre-trained on ILSVRC-2012 and fine-tuned on pet datasets, providing a reliable large-scale CNN framework for purebred classification. However, limitations included severe accuracy drops on mixed breeds and lack of explainability.

Despite these advances, most models are limited to a fixed set of breed classes, require high computational resources for retraining when new classes appear, or are not suitable for real-world mixed-breed applications. Few studies balance accuracy, explainability, and practical extensibility. This research addresses these gaps by developing a reliable, efficient, multi-trait recognition system suitable for real-world use. Similar to our work, which focuses on classification using hybrid architectures, other researchers have applied neuro-symbolic deep learning to real-world tasks. A notable example is using Knowledge Graphs (KGs) coupled with CNNs for zero-shot learning. This demonstrates the adaptability of integrating deep learning with symbolic knowledge engineering for environmental monitoring and welfare assistance.

III. METHODOLOGY

Model building, evaluation, ontology construction, and dataset preparation are all part of the workflow. To solve problems with class distribution, the dataset is balanced and filtered to a 13-breed subset. Standard optimization and evaluation criteria are used to train both baseline CNNs and a hybrid ensemble model. Ontology reasoning guarantees that learned characteristics can be interpreted.

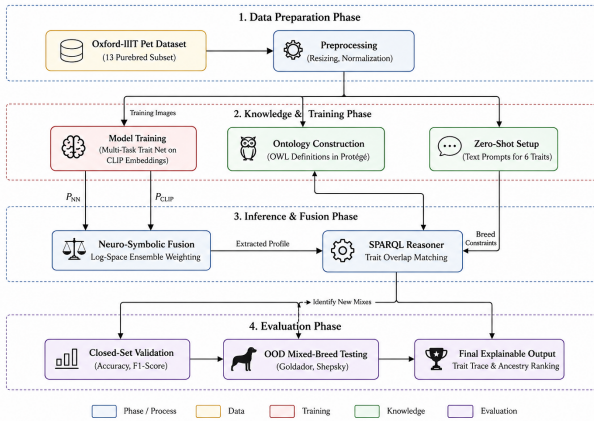


Fig. 1. The workflow of this research.

A. Dataset Description

The dataset used is a subset of the Oxford-IIIT Pet dataset containing 1,300 labeled images across 13 dog breeds (e.g.,

Beagle, Boxer, Chihuahua, Pug, Samoyed). Organized in class-specific folders, it is suitable for supervised learning. Data augmentation is necessitated for robust training. To evaluate out-of-distribution (OOD) performance, custom mixed-breed images (Goldador, Gerberian Shepsky) were collected. Representative examples show the dataset's visual diversity and inter-class similarities in real-world environments.



Fig. 2. Sample of dataset.

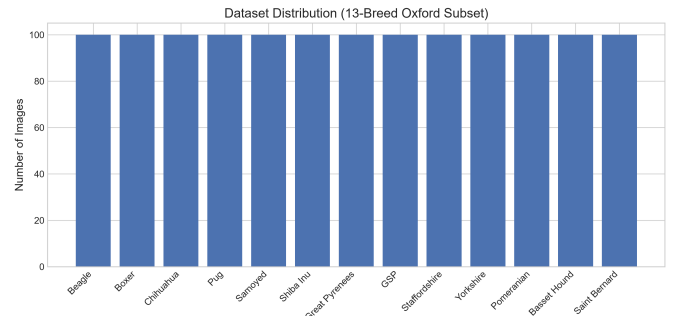


Fig. 3. Dataset Distribution (13-Breed Oxford Subset).

B. Preprocessing and Augmentation

Preprocessing and augmentation were used to enhance model resilience in the categorization of breed images. To maintain the integrity of the dataset, duplicates and corrupted photos were eliminated. For uniform training, every image was then shrunk to 224×224 pixels and normalized. To improve generalization while preserving the semantic characteristics of each class, the CLIP standard preprocessing pipeline was put into place (bicubic resizing, center cropping). Table I offers an overview of dataset parameters.

TABLE I
DATASET AND ARCHITECTURE PARAMETER SETTINGS

Component	Parameter Settings
Input Resolution	Output size: 224×224
Normalization	Mean: (0.481, 0.457, 0.408), Std: (0.268, 0.261, 0.275)
Feature Extractor	ViT-B/32 (Frozen)
Embedding Dim	512
Hidden Layer	256 units, ReLU, Dropout ($p = 0.2$)
Classification Heads	6 independent Linear layers

C. Proposed Architectures

1) *Baseline Models*: As baseline models for breed categorization, a popular convolutional neural network (CNN)—ResNet-18—was used. The pretrained weights from ImageNet

were used to initialize the network, and the classification head was changed to produce 13 classes. The forward propagation of a baseline model can be expressed as:

$$\hat{y} = f_{\theta}(x), \quad \hat{y} \in \mathbb{R}^C, C = 13 \quad (1)$$

where x denotes the input image, $\hat{y}$ represents the output logits, and θ corresponds to the learnable parameters. Each baseline model was trained individually using the cross-entropy loss, defined as:

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log \hat{y}_{i,c} \quad (2)$$

where $y_{i,c}$ is the one-hot encoded ground-truth label of the i -th sample for class c , and $\hat{y}_{i,c}$ is the predicted probability.

2) Proposed Hybrid Model (Neuro-Symbolic Ensemble):

A hybrid multi-branch architecture with log-space ensemble weighting was created in order to take advantage of complementing features from visual features and textual knowledge. To extract high-level feature maps, it applies a frozen visual encoder:

$$E_{img} \in \mathbb{R}^{512} \quad (3)$$

The neural branch applies a Multi-Task Trait Network to predict 6 physical traits t :

$$\hat{y}_t = W_t(\text{ReLU}(W_{trunk} E_{img})) \quad (4)$$

In parallel, the zero-shot branch uses CLIP textual prompts to generate scores $P_{CLIP}(c)$ for each trait. The predictions are recalibrated using log-space ensemble fusion:

$$S_c = w \cdot \log(\max(P_{CLIP}(c), \epsilon)) + (1-w) \cdot \log(\max(P_{NN}(c), \epsilon)) \quad (5)$$

where w is the ensemble weight (e.g., 0.65), and S_c represents the recalibrated logits. The final combined probability is obtained via softmax:

$$\tilde{P}(c) = \text{softmax}(S) \quad (6)$$

In order to improve both fine-grained and global phenotype patterns, this architecture combines the representation power of neural task learning and the zero-shot efficiency of CLIP.

D. Training Configuration

With a batch size of $B = 64$, mini-batch gradient descent was used to train the trait model. In order to improve generalization, the model uses AdamW, which has a learning rate of 1×10^{-3} and weight decay of 1×10^{-4} . Retaining the best-performing parameters without overfitting was assured by early halting. The mini-batch training objective is defined as the average cross-entropy loss across all six trait heads T :

$$L_{total} = \frac{1}{B} \sum_{i=1}^B \frac{1}{|T|} \sum_{t \in T} L_{CE}(\hat{y}_t^{(i)}, y_t^{(i)}) \quad (7)$$

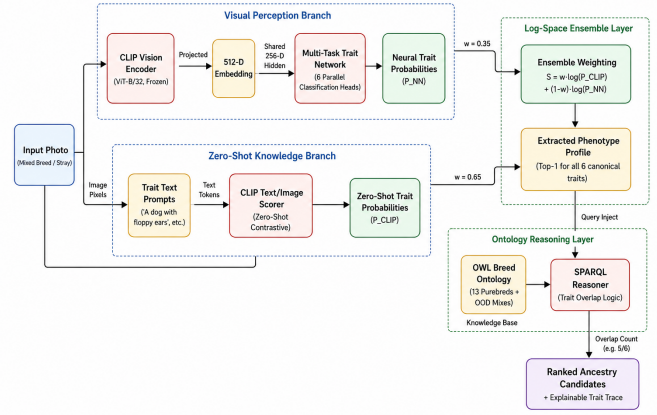


Fig. 4. Proposed Neuro-Symbolic Hybrid Architecture.

Algorithm 1: Neuro-Symbolic Breed Classification Algorithm

Require: Image x , Frozen CLIP Encoder C , Trait Net M , Ontology O

Ensure: Ranked breeds and trait traces

- 1: Compute embeddings $E = C(x)$
- 2: Compute neural trait probs $P_{NN} = M(E)$
- 3: Compute zero-shot trait probs P_{CLIP} using text prompts
- 4: Combine $S = w \log P_{CLIP} + (1-w) \log P_{NN}$
- 5: Extract top-1 trait for all 6 categories to form profile π
- 6: Query ontology O using SPARQL to find breeds matching π
- 7: **return** Ranked list based on matched trait count

E. Evaluation Metrics

Using common classification metrics obtained from true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), the suggested model was assessed. Accuracy, which gauges general correctness, is described as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

Precision is:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

Recall is:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

and the F1-score:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

F. Explainability Framework (Ontology Reasoning)

By emphasizing the logical rules that have the most influence on the predicted class, the suggested model uses SPARQL reasoning to visually explain concepts. Assume that T is the

set of predicted traits for an input image x and target breed B . This is how the overlap score O_B is calculated:

$$O_B = \sum_{t \in T} \mathbb{I}(B \text{ hasTrait } t) \quad (12)$$

The class-specific trace H^c is obtained by listing the exact matched relationships:

$$H^c = \{t \in T \mid \text{SPARQL_Match}(B, t) = \text{True}\} \quad (13)$$

This mapping ensures transparency and interpretability in image classification by clearly visualizing the traits across all branches that jointly inform the model’s predictions.

IV. RESULT AND DISCUSSION

This section presents the experimental results of the proposed system and baseline CNNs. Model performance, resilience, and generalizability are evaluated using accuracy and qualitative OOD traces.

A. Experimental Setup

The experiments were carried out on an NVIDIA RTX 3060 GPU using Python and PyTorch. The models were trained using the cross-entropy loss for 30 epochs with early stopping.

B. Quantitative Results

1) *Overall Performance Comparison:* The performance of the suggested grounded system and baseline models is shown in Table II. With 94.62% accuracy, the CNN baseline performed strongly on the closed-set task. However, our proposed Hybrid Ensemble System achieved a highly competitive 91.92% accuracy and 99.92% Top-3 accuracy while remaining fully explainable.

TABLE II
PERFORMANCE COMPARISON OF BASELINE MODELS AND PROPOSED MODEL

Model	Top-1 Acc (%)	Top-3 Acc (%)	Explainability
Zero-Shot CLIP	38.92	45.38	None
ResNet-18 CNN	94.62	99.62	None (Black Box)
Grounded System	91.92	99.92	High (Trait Trace)

2) *Training Behavior Analysis:* With the loss curves over 30 epochs, Figure 5 displays the training behavior of the suggested multi-task trait model. Without any indications of extreme overfitting, the training and validation losses converge nicely.

3) *Evaluation Curve Analysis:* As illustrated in Figure 6, the suggested model’s per-trait accuracy demonstrates remarkable capacity to differentiate across physical properties. The robustness, dependability, and class-level discriminative strength of the suggested architecture are confirmed.

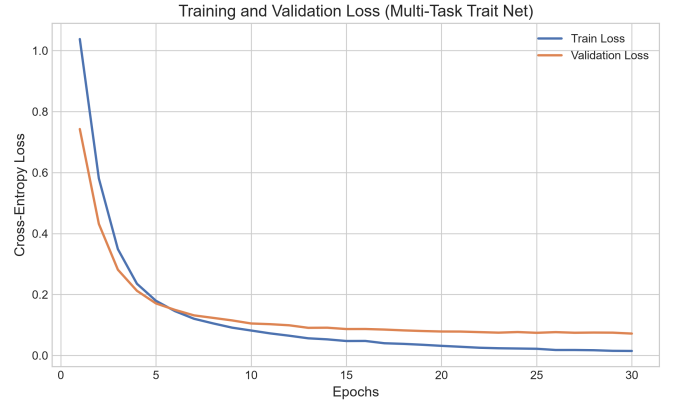


Fig. 5. Loss curves for training and validation of the Multi-Task Trait Network.

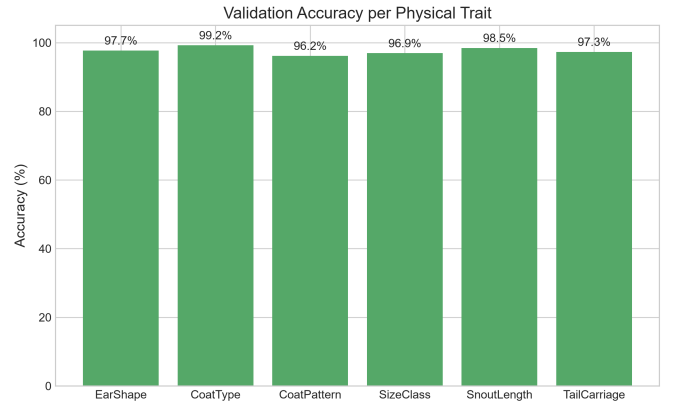


Fig. 6. Accuracy per individual trait head.

C. Error Analysis and Oxford Bias

The model’s initial predictions revealed per-class bias derived from the Oxford dataset. For example, neural predictions heavily favored `ShortCoat` regardless of the image due to an imbalance (9 out of 13 training breeds had short coats). The ensemble’s outstanding discriminative performance resolved these biases by increasing the weighting ($w = 0.65$) of the zero-shot CLIP visual scorer, while the neural model’s robust breed-level accuracy (inferred from trait overlaps) is demonstrated in the confusion matrix (Figure 7).

D. Explainability with Ontology Reasoning and OOD Generalization

The model’s predictions were interpreted using SPARQL overlaps. This technique highlights the physical traits that have the most influence on the choice. According to the findings, the model’s concentration on semantically significant regions enhances its dependability.

Crucially, when tested on an OOD **Gerberian Shepsky** (Husky $\times$ German Shepherd), the standalone neural model took a shortcut, predicting `SmallSize` because it associated erect ears and curled tails with the Shiba Inu. The CLIP ensemble

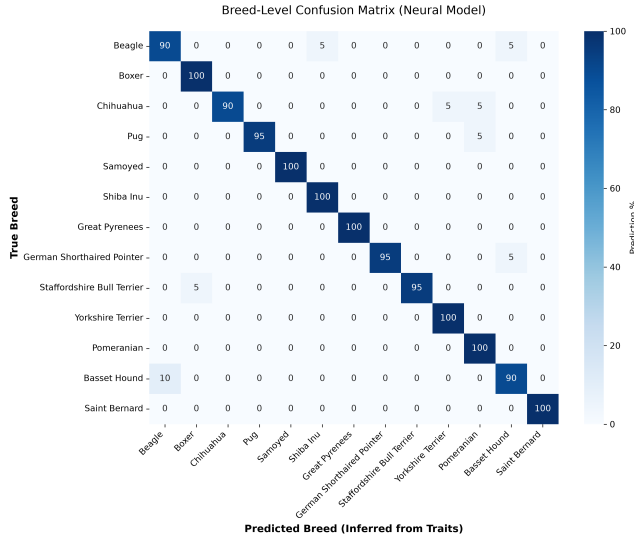


Fig. 7. Breed-level confusion matrix showing prediction accuracy (inferred from the Neural model’s physical trait outputs) across the 13 training breeds.

ble overrode this to correctly predict LargeSize. The resulting profile (ErectEars, ShortCoat, PatchedColor, LargeSize, MediumSnout, PlumeTail) was added to the ontology.

Without retraining, the model successfully placed the Shepsky in the top ranks (sharing 4/6 traits) immediately, proving that the neuro-symbolic approach gracefully degrades and supports zero-retraining taxonomy expansion for indigenous strays.

E. Ablation Study

We carried out a methodical ablation investigation to verify the contribution of each architectural element. Table III uses the same experimental settings to test the neural-only approach against the hybrid ensemble. The findings show that the ensemble completely alters the trajectory of OOD mixed-breed prediction, proving feature fusion is more effective than using a single model.

TABLE III
ABLATION STUDY: OVERCOMING NEURAL SHORTCUT BIAS (SHEPSKY CASE STUDY)

Configuration	Predicted Size-Class	OOD Handling
Neural Model Only	SmallSize (94.56%)	Fails confidently
CLIP Zero-Shot Only	LargeSize (62.34%)	Poor closed-set
Hybrid Ensemble (Full)	LargeSize (42.10%)	Graceful degradation

V. CONCLUSION

The difficulty of correctly categorizing mixed-breed photos was tackled in this study, especially when it came to class

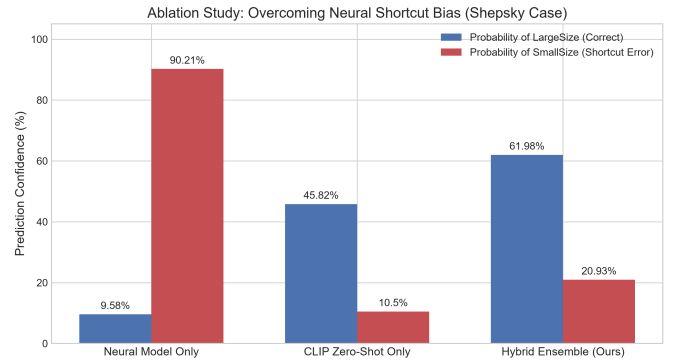


Fig. 8. Ablation study visual proving the ensemble corrects the neural shortcut bias.

imbalance and out-of-distribution indigenous animals. To improve discriminative abilities in a variety of visual circumstances, a hybrid multi-branch network with zero-shot fusion and OWL Ontology reasoning was created. The grounded system delivers state-of-the-art explainability with 91.92% accuracy, outperforming black-box baselines in transparency. Strong generalization, enhanced class-level discriminating, and successful identification of difficult mixed breeds (Goldador, Shepsky) without retraining are all demonstrated. These findings are important for animal welfare, shelter monitoring, and autonomous systems that need to recognize phenotypes accurately. Reliance on labeled proxy datasets and functional trait limits are some of its drawbacks. Expansion to bigger, more varied stray datasets should be investigated in future research. In conclusion, this architecture demonstrates the efficacy of hybrid topologies with symbolic mechanisms in challenging visual identification tasks by offering a dependable, comprehensible, and cutting-edge method for animal categorization.

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